

4. Form the V matrix using the v_i vectors as rows; see Equation (6.132).
5. Use $\hat{D}$ inner products to form the unit upper triangular matrix U^- ; see Equations (6.129) and (6.132).
6. Define D^- as $D^- = V\hat{D}V^T$.

The U-D filter results in twice as much precision as the standard Kalman filter, just like the square root filter, but it requires less computation than the square root filter. If some of the states are missing from the measurement vector, a more efficient U-D algorithm can be derived [Bar83].

6.5 SUMMARY

In this chapter, we discussed the sequential Kalman filter, which is mathematically identical to the Kalman filter, but which avoids matrix inversion. This is an attractive formulation for embedded systems in which computational time and memory are at a premium. However, sequential filtering can only be used if the noise covariance is diagonal, or if the noise covariance is constant. Information filtering is also equivalent to the Kalman filter, but it propagates the inverse of the covariance. This can be computationally beneficial in cases in which the number of measurements is much larger than the number of states. Square root filtering and U-D filtering effectively increase the precision of the Kalman filter. Although these approaches require additional computational effort, they can help prevent divergence and instability. Gerald Bierman's book provides an excellent and comprehensive overview of square root and U-D filtering [Bie77b].

We see that we have a number of different choices when implementing a Kalman filter.

- Covariance filtering or information filtering
- Standard filtering, square root filtering, or U-D filtering
- Batch filtering or sequential filtering

Any of these choices can be made independently of the other choices. For instance, we can choose to combine information filtering with square root filtering [Kam71] in much the same way as we combined covariance filtering with square root filtering in this chapter. The choices in the list above gives us a total of 12 different Kalman filter formulations (two choices in the first item, three choices in the second item, and two choices in the third item). There are also other choices that are not listed above, especially other types of square root filtering. A numerical comparison of various Kalman filter formulations (including the standard filter, the square root covariance filter, the square root information filter, and the Chandrasekhar algorithm) is given in [Ver86]. Numerical and computational comparisons of various Kalman filtering approaches are given in [Bie73, Bie77a]. Continuous-time square root filtering is discussed in [Mor78] and in Section 8.3.3 of this book.

PROBLEMS

Written exercises

6.1 In this chapter, we discussed alternatives to the standard Kalman filter formulation. Some of these alternatives include the sequential Kalman filter, the information filter, and the square root filter.

- What is the advantage of the sequential Kalman filter over the batch Kalman filter? What is the advantage of the batch Kalman filter over the sequential Kalman filter?
- What is the advantage of the information filter over the standard Kalman filter? What is the advantage of the standard Kalman filter over the information filter?
- What is the advantage of the square root filter over the standard Kalman filter? What is an advantage of the standard Kalman filter over the square root Kalman filter?

6.2 Suppose that you have a system with the following measurement and measurement noise covariance matrices:

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

You want to use a sequential Kalman filter to estimate the state of the system. Derive the normalized measurement, measurement matrix, and measurement noise covariance matrix that could be used in a sequential Kalman filter.

6.3 Consider the two alternative forms for the information matrix time-update equation. What advantages does Equation (6.28) have? What advantages does Equation (6.30) have?

6.4 A radioactive mass has a half-life of τ seconds. At each time step k the number of emitted particles x is half of what it was one time step ago, but there is some error w_k (zero-mean with variance Q_k) in the number of emitted particles due to background radiation. At each time step the number of emitted particles is counted with two separate and independent instruments. The instruments used to count the number of emitted particles both have a random error at each time step that is zero-mean with a unity variance. The initial uncertainty in the number of radioactive particles is a random variable with zero mean and unity variance.

- The discrete-time equations that model this system have a one-dimensional state and a two-dimensional measurement. Use the information filter to compute the *a priori* and *a posteriori* information matrix at $k = 1$ and $k = 2$. Assume that $Q_0 = 1$ and $Q_1 = 5/4$.
- Another way to solve this problem is to realize that the two measurements can be averaged to form a single measurement with a smaller variance than the two independent measurements. What is the variance of the averaged measurement at each time step? Use the standard Kalman filter equations

to compute the *a priori* and *a posteriori* covariance matrix at $k = 1$ and $k = 2$, and verify that it is the inverse of the information matrix that you computed in part (a).

6.5 Prove that the singular values of a diagonal matrix are the magnitudes of the diagonal elements.

6.6 Prove that SS^T is symmetric positive semidefinite for any S matrix.

6.7 Find an upper triangular matrix S (using only paper and pencil) such that

$$SS^T = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$$

Is your solution unique?

6.8 Find an upper triangular matrix S (using only paper and pencil) such that

$$SS^T = \begin{bmatrix} 5 & 2 & -2 \\ 2 & 2 & -1 \\ -2 & -1 & 1 \end{bmatrix}$$

How many solutions exist to this problem?

6.9 Verify Equation (6.70). Hint: Equate the two sides of the equation, take the trace, and solve for γ . Make sure to explain why taking the trace is valid.

6.10 Suppose that an orthogonal matrix $\tilde{T}$ is desired to satisfy Equation (6.97), where Cholesky factorization is used to compute the matrix square roots on the left side of the equation. This equation can then be written as $U = \tilde{T}A$, where U is an upper triangular matrix. Show that such a transformation cannot be found unless the two-norm of the first column of A happens to be equal to $|U_{11}|$. [Note that this does not necessarily prevent the possibility of the transformation of Equation (6.97), because U could be nontriangular if nontriangular square root matrices are used to form the U matrix.]

6.11 Use the Householder method (using only paper and pencil) to find an orthogonal T such that $TA = \begin{bmatrix} W \\ 0 \end{bmatrix}$ where W is a 2×2 matrix and

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 1 \\ 2 & 2 \end{bmatrix}$$

6.12 Use the modified Gram-Schmidt method (using only paper and pencil) to solve Problem 6.11.

6.13 Compute the U-D factorization (using only paper and pencil) for the matrix

$$P = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$$

Computer exercises

6.14 Consider the RLC circuit of Example 1.8 with $R = 100$ and $L = C = 1$. Suppose the applied voltage is continuous-time, zero-mean white noise with a standard deviation of 3. The initial capacitor voltage and inductor current are both zero. Discretize the system with a time step of 0.1. The discrete-time measurements consist of the capacitor voltage and the inductor current, both measurements containing zero-mean unity variance noise. Implement a sequential Kalman filter for the system. Simulate the system for 2 seconds. Let the initial state estimate be equal to the initial state, and the initial estimation covariance be equal to $0.1I$. Hint: Set the discrete-time process noise covariance $Q = Q_c \Delta t$, where Q_c is the covariance of the continuous-time process noise, and Δt is the discretization step size. Q will be nondiagonal, which means you need to use the algorithm in Section 2.7 to simulate the process noise.

- Generate a plot showing the *a priori* variance of the capacitor voltage estimation error, and the two *a posteriori* variances of the capacitor voltage estimation error.
- Generate a plot showing a typical trace of the true, *a posteriori* estimated, and measured capacitor voltage. What is the standard deviation of the capacitor voltage measurement error? What is the standard deviation of the capacitor voltage estimation error?

6.15 The pitch motion of an aircraft flying at constant speed can be approximately described by the following equations [Ste94]:

$$\begin{aligned}\dot{x} &= \begin{bmatrix} -0.5680 & 17.9800 \\ 1.0000 & -1.2370 \end{bmatrix} x + \begin{bmatrix} 0.1750 & 0.1750 \\ -0.0010 & -0.0010 \end{bmatrix} u + \begin{bmatrix} 17.9800 \\ -1.2370 \end{bmatrix} w \\ y(t_k) &= x(t_k) + v_k\end{aligned}$$

where x_1 is the pitch rate, x_2 is the angle of attack, u consists of the elevator and flap angles, and w is disturbance due to wind. Suppose that the variance of the wind disturbance is 0.001, and the measurement variances are 0.3. Discretize the system with a step size of 0.01 and simulate the system and a square root Kalman filter for 100 time steps. Use an initial state of zero, an initial state estimate of zero, an initial estimation-error covariance of $0.01I$, and a control input of zero. Hint: Set the discrete-time process noise covariance $Q = Q_c \Delta t$, where Q_c is the covariance of the continuous-time process noise, and Δt is the discretization step size. Q will be nondiagonal, which means you need to use the algorithm in Section 2.7 to simulate the process noise.

- Generate a plot showing the *a posteriori* variance of the estimation errors of the two states.
- Generate a plot showing a typical trace of the true, *a posteriori* estimated, and measured pitch rate. What is the standard deviation of the pitch rate measurement error? What is the standard deviation of the pitch rate estimation error?
- Generate a plot showing a typical trace of the true, *a posteriori* estimated, and measured angle of attack. What is the standard deviation of the angle of attack measurement error? What is the standard deviation of the angle of attack estimation error?

